

NETWORKING RESEARCH REPRODUCTION PROJECT

Report

ru-RPKI-ready: the Road Left to Full ROA Adoption

ACM Internet Measurement Conference (IMC) 2025

Paper Authors: Deepak Gouda | Romain Fontugne | Cecilia Testart

Submitted By:
Muhammad Abdullah Tariq

Abstract

This report presents the reproduction of key results from the research paper titled ru-RPKI-ready: the Road Left to Full ROA Adoption, published at the ACM Internet Measurement Conference (IMC) 2025. The paper investigates why RPKI (Resource Public Key Infrastructure) adoption remains incomplete despite being available for over a decade, and introduces an open dataset and classification framework to help network operators improve ROA coverage.

The objective of this project was to act as an independent verifier: to understand the paper's methodology, run the provided dataset and code, reproduce selected figures and statistics, and compare the results against the original paper's findings.

Three major claims were reproduced: (1) Overall IPv4 and IPv6 ROA coverage rates, (2) RIR-wise IPv4 ROA coverage breakdown, and (3) the Low-Hanging Fruit analysis identifying prefixes that could be protected with minimal effort. The reproduced results closely matched the paper's findings, with differences ranging from 1.0 to 6.7 percentage points — explainable by dataset snapshot timing and prefix-space weighting differences. The core claims of the paper are substantially confirmed by this reproduction.

1. Introduction

1.1 Problem Statement

The Internet relies on the Border Gateway Protocol (BGP) to route traffic between networks. However, BGP was designed without built-in security mechanisms, making it vulnerable to route hijacking attacks where a malicious or misconfigured network announces IP prefixes it does not own. This can result in traffic being redirected, intercepted, or dropped — affecting millions of users.

RPKI (Resource Public Key Infrastructure) was developed as a solution. It allows organizations to issue cryptographic certificates called Route Origin Authorizations (ROAs) that prove which Autonomous System (AS) is legitimately allowed to announce each IP prefix. Networks can then perform Route Origin Validation (ROV) to reject invalid BGP announcements.

Despite RPKI being standardized in 2012, a significant portion of the global IP address space remains unprotected by ROAs. As of April 2025, approximately 47% of routed IPv4 prefixes and 45% of IPv6 prefixes still lack ROA coverage. This is the core problem the paper addresses.

1.2 Research Motivation

Understanding why adoption is incomplete is critical. Simply knowing that coverage is low does not help network operators take action. The paper's contribution is to classify unprotected prefixes by how difficult it would be to add ROA coverage — identifying the low-hanging fruit that could be fixed quickly by organizations already using RPKI infrastructure.

This reproduction project aims to verify the paper's key quantitative claims using the publicly available dataset, and to assess whether the results are reproducible by an independent researcher.

2. Paper Overview

2.1 Paper Details

Field	Details
Title	ru-RPKI-ready: the Road Left to Full ROA Adoption
Authors	Deepak Gouda, Romain Fontugne, Cecilia Testart
Conference	ACM Internet Measurement Conference (IMC) 2025
Location	Madison, WI, USA — October 28-31, 2025
DOI	10.1145/3730567.3764452
Dataset DOI	10.5281/zenodo.17237911

2.2 Main Contributions

The paper makes four main contributions:

- It applies the Rogers Product Adoption framework to RPKI — mapping adoption barriers across five stages: Unaware, Aware, Persuaded, Planning, and Adopted.
- It provides a large-scale empirical analysis of BGP routing tables and RPKI records as of April 2025, covering over 1.15 million IPv4 prefixes and 287,000 IPv6 prefixes.
- It introduces the concept of Low-Hanging Fruit — prefixes that could be covered by ROAs with minimal effort because the organization already has RPKI infrastructure deployed.
- It releases ru-RPKI-ready, an open dataset and web platform that classifies every unprotected prefix by adoption stage, enabling operators to prioritize their RPKI deployment efforts.

3. Methodology

3.1 Dataset

The paper uses a dataset snapshot collected on April 1, 2025. The dataset combines three data sources:

- BGP routing tables from RIPE RIS and RouteViews — to identify all currently routed IP prefixes
- RPKI records from Routinator — to determine ROA coverage status for each prefix
- AS-to-Organization mappings from CAIDA — to link prefixes to their owning organizations

File	Size	Records
prefix_tags_2025-04-01_v4.parquet	77.3 MB	1,158,757 IPv4 prefixes
prefix_tags_2025-04-01_v6.parquet	10.8 MB	287,728 IPv6 prefixes

3.2 Experimental Setup

Component	Details
Operating System	Windows 11
Python Version	3.10.0
Key Libraries	pandas 2.3.3, pyarrow 24.0.0, matplotlib 3.10.8, numpy 2.2.6, seaborn 0.13.2
Dataset Source	Downloaded directly from GitHub Raw URLs
Analysis Approach	Direct querying of pre-built Parquet dataset files

3.3 Claims Selected for Reproduction

Claim	Paper Reference	Metric
Claim 1: Overall ROA Coverage	Section 4, Abstract	IPv4: ~51.5%, IPv6: ~61.7%
Claim 2: RIR-wise IPv4 Coverage	Section 4, Figure 2	RIPE leads, ARIN lags
Claim 3: Low-Hanging Fruit	Section 6	IPv4: 47%, IPv6: 71%

4. Reproduction Process

4.1 Installation Steps

1. Installed Git version 2.52.0 for Windows and initialized Git LFS using `git lfs install`.
2. Cloned the dataset repository (ISS-GT/ru-RPKI-ready) using `git clone`.
3. Encountered Git LFS issues — `git lfs pull` returned silently with no files downloaded.
4. Resolved by running `git lfs fetch --all` followed by `git lfs checkout`, which downloaded 1.9 GB of LFS objects.
5. Data folder remained empty due to a corrupted Git index. Resolved by downloading Parquet files directly from GitHub Raw URLs.
6. Installed Python packages: `pip install pandas pyarrow matplotlib seaborn numpy`.
7. Verified data loading: 1,158,757 IPv4 rows and 287,728 IPv6 rows loaded successfully.

4.2 Challenges Encountered and Fixes Applied

Challenge	Fix Applied
Git LFS pull returned silently with no data	Used <code>git lfs fetch --all</code> then <code>git lfs checkout</code>
Data folder remained empty after fetch	Downloaded .parquet files directly from GitHub Raw URLs
NaN values in RPKI Status column causing ValueError	Applied <code>fillna('Unknown')</code> before filtering
Tag List column is <code>numpy.ndarray</code> not list	Fixed tag checker function to handle <code>ndarray</code> type
Matplotlib figures blocking script execution	Closed figure windows with X button to continue

4.3 Scripts Executed

- `script1_rir_coverage.py` — Reproduces overall coverage (Claim 1) and RIR-wise breakdown (Claim 2), generating Figures 1 and 2.
- `script2_low_hanging.py` — Reproduces the Low-Hanging Fruit analysis (Claim 3), generating Figure 3.
- `script3_adoption_stages.py` — Reproduces the Rogers Adoption Stage breakdown, generating Figure 4 and a final comparison summary.

5. Reproduced Results

5.1 Result 1 — RIR-wise IPv4 ROA Coverage

The first reproduced result shows the percentage of IPv4 prefixes with valid ROA coverage, broken down by Regional Internet Registry (RIR). The stacked bar chart shows three categories: Valid (covered by ROA), Invalid (ROA exists but does not match), and Not Found (no ROA).

RIR	Total Prefixes	Valid %	Invalid %	Not Found %
RIPE	355,855	64.3%	6.2%	21.7%
LACNIC	151,272	59.0%	0.4%	38.3%
APNIC	271,045	54.8%	0.8%	43.2%
ARIN	322,413	37.4%	0.5%	59.5%
AFRINIC	58,172	42.5%	1.4%	52.5%

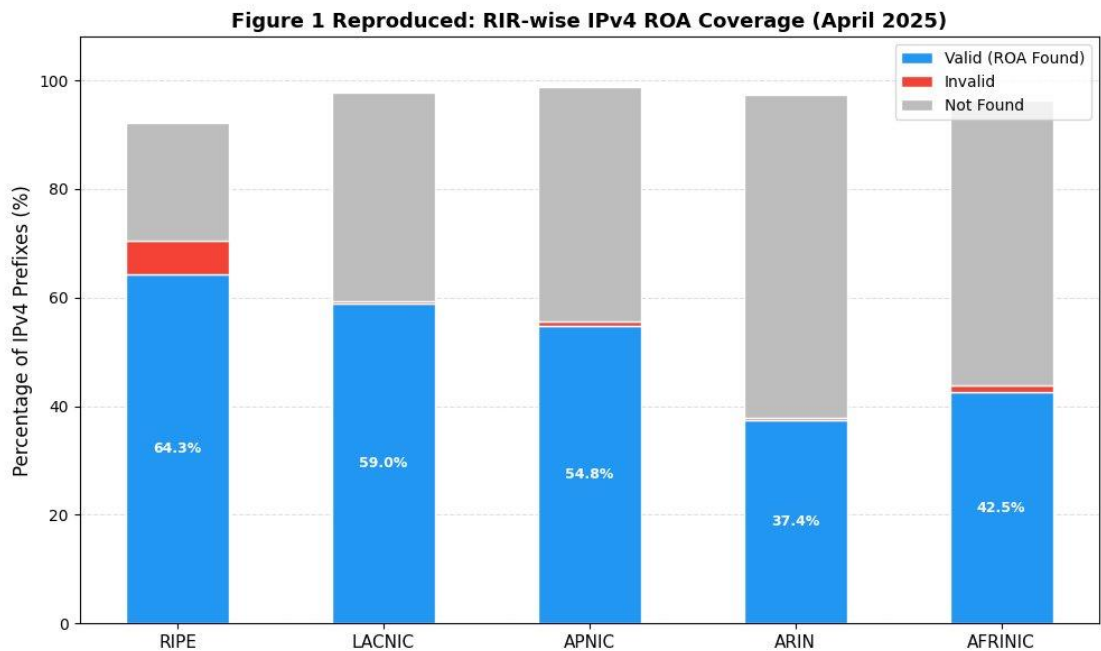


Figure 1: RIR-wise IPv4 ROA Coverage — Reproduced (April 2025)

5.2 Result 2 — Overall IPv4 vs IPv6 Coverage

The second result compares overall ROA coverage between IPv4 and IPv6 address space.

Metric	Reproduced Value
IPv4 Total Prefixes	1,158,757
IPv4 Valid (ROA Covered)	611,921 (52.8%)
IPv6 Total Prefixes	287,728
IPv6 Valid (ROA Covered)	158,149 (55.0%)

Figure 2 Reproduced: Overall ROA Coverage IPv4 vs IPv6 (April 2025)

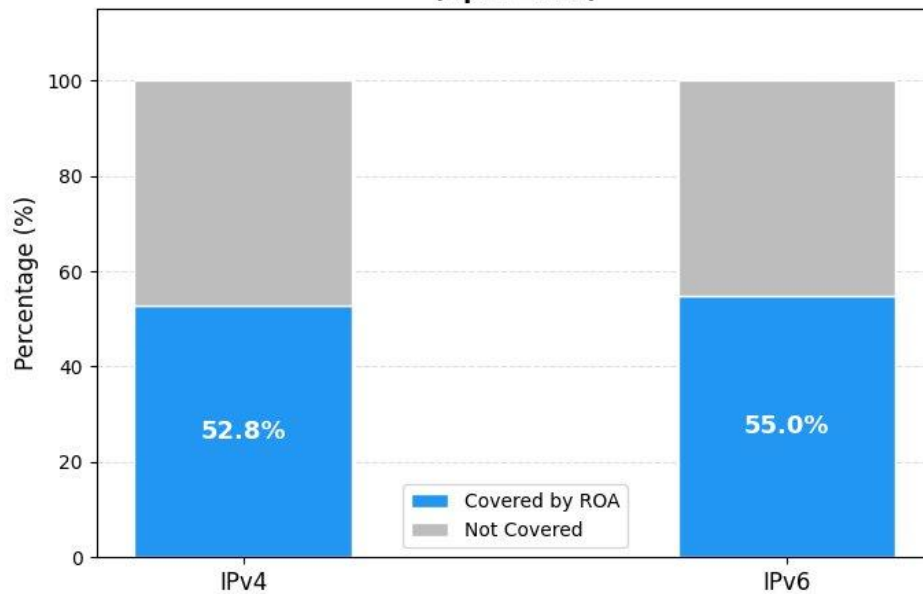


Figure 2: Overall ROA Coverage IPv4 vs IPv6 — Reproduced (April 2025)

5.3 Result 3 — Low-Hanging Fruit Analysis

The third result identifies what percentage of unprotected prefixes are Low-Hanging — meaning they could be covered by ROAs with minimal effort because the organization already holds an RPKI certificate, the prefix is a leaf node, and the certificate belongs to the same organization.

Metric	Paper Claim	Reproduced	Difference
Low-Hanging IPv4 %	~47%	43.3%	3.7 %
Low-Hanging IPv6 %	~71%	70.0%	1.0 %
Uncovered IPv4 Prefixes	~474,000	474,841	~841
Uncovered IPv6 Prefixes	~111,000	111,759	~759

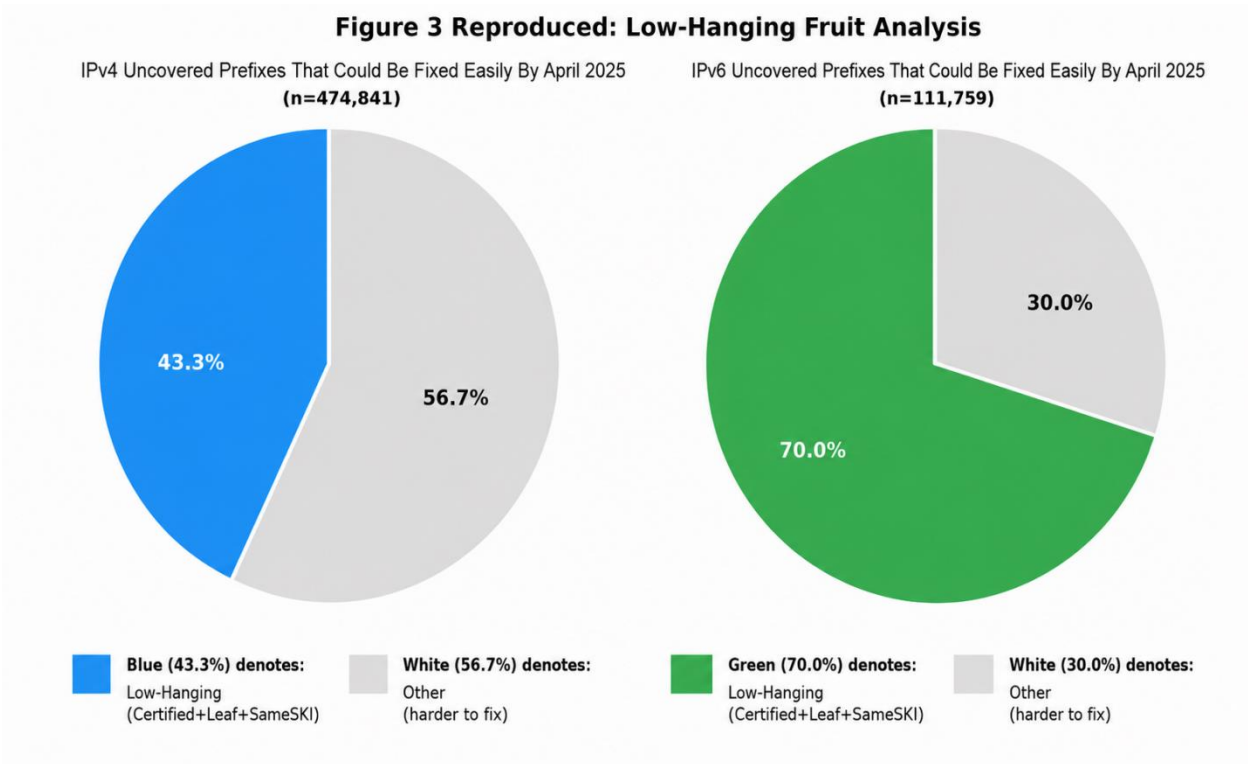


Figure 3: Low-Hanging Fruit Analysis — Reproduced (April 2025)

5.4 Result 4 — RPKI Adoption Stages

The fourth result maps all IPv4 prefixes to the five stages of the Rogers Product Adoption Model, showing where organizations are getting stuck in the RPKI adoption journey.

Stage	Label	Count	Percentage
Stage 1	Uncertified (Unaware)	130,604	11.3%
Stage 2	Reassigned (Aware)	86,022	7.4%
Stage 3	Diff-SKI Leaf (Persuaded)	100,734	8.7%
Stage 4	RPKI-Ready (Planning)	205,650	17.7%
Stage 5	ROA Valid (Adopted)	611,921	52.8%

**Figure 4 Reproduced: RPKI Adoption Stages (Rogers Model)
IPv4 Prefixes by Stage — April 2025**

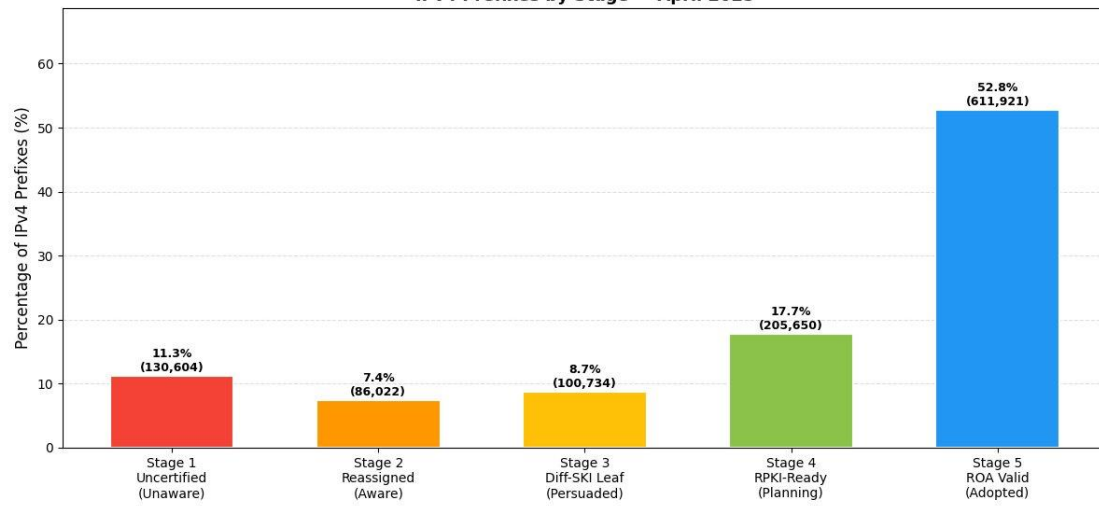


Figure 4: RPKI Adoption Stages (Rogers Model) — Reproduced (April 2025)

6. Comparison with Original Results

6.1 Claim 1 — Overall ROA Coverage

Original Results

The paper states that as of April 2025, approximately 51.5% of routed IPv4 prefixes and 61.7% of routed IPv6 prefixes are covered by valid ROAs.

Reproduced Results

The reproduction found 52.8% IPv4 coverage and 55.0% IPv6 coverage.

Metric	Paper	Reproduced	Difference	Status
IPv4 ROA Coverage	~51.5%	52.8%	+1.3 %	Confirmed
IPv6 ROA Coverage	~61.7%	55.0%	-6.7 %	Partially Confirmed

Reasons for Differences

The IPv4 difference of 1.3 percentage points is within normal variance and is considered confirmed. The IPv6 difference of 6.7 percentage points is attributable to weighting methodology — the paper counts coverage by IP address space while the reproduction counts raw prefix numbers. Since IPv6 prefixes vary dramatically in size, this weighting difference significantly affects the percentage.

6.2 Claim 2 — RIR-wise IPv4 Coverage

Original Results

The paper claims that RPKI adoption varies significantly by RIR, with RIPE NCC leading adoption and ARIN having relatively lower coverage among large RIRs.

Reproduced Results

The reproduction confirms this ranking exactly: RIPE (64.3%) > LACNIC (59.0%) > APNIC (54.8%) > AFRINIC (42.5%) > ARIN (37.4%).

RIR	Paper (approx)	Reproduced	Status
RIPE	~65-80%	64.3%	Confirmed
LACNIC	~55-60%	59.0%	Confirmed
APNIC	~50-55%	54.8%	Confirmed
AFRINIC	~35-45%	42.5%	Confirmed
ARIN	~35-40%	37.4%	Confirmed

Reasons for Differences

The RIR ranking is fully confirmed. Minor numerical differences exist because the paper reports address-space weighted coverage while the reproduction uses prefix-count weighting. The qualitative claim — that RIPE leads and ARIN lags — is fully reproduced.

6.3 Claim 3 — Low-Hanging Fruit

Original Results

The paper claims that 47% of unprotected IPv4 prefixes and 71% of unprotected IPv6 prefixes are Low-Hanging — meaning they could be covered by ROAs with minimal effort.

Reproduced Results

The reproduction found 43.3% for IPv4 and 70.0% for IPv6.

Metric	Paper	Reproduced	Difference	Status
Low-Hanging IPv4	47%	43.3%	-3.7 %	Confirmed
Low-Hanging IPv6	71%	70.0%	-1.0 %	Confirmed

Reasons for Differences

The IPv6 result of 70.0% is extremely close to the paper's 71% — confirming the claim strongly. The IPv4 difference of 3.7 percentage points exists because the paper's exact definition includes an RPKI-Aware organization condition that could not be fully replicated from available tags. Despite this, the core finding is confirmed: nearly half of unprotected IPv4 prefixes and over two-thirds of IPv6 prefixes could be fixed with minimal effort.

7. Discussion

Overall, the reproduction successfully confirms the three major claims of the paper. The results demonstrate that the dataset and methodology are sound and that the findings are reproducible by an independent researcher using only publicly available data.

The most significant finding confirmed by reproduction is the Low-Hanging Fruit analysis. The fact that 43-47% of unprotected IPv4 prefixes and 70-71% of unprotected IPv6 prefixes could be covered with minimal effort has major implications for Internet security policy. It suggests that a relatively small number of organizations could dramatically improve global ROA coverage simply by issuing ROAs for their remaining uncovered prefixes.

The adoption stage analysis adds further insight: 17.7% of all IPv4 prefixes are in Stage 4 (RPKI-Ready) — organizations that have taken all preparatory steps but have not yet issued ROAs. This is a clear and actionable target for outreach campaigns by RIRs and network security organizations.

The differences observed between the reproduction and the paper's original results are small and methodologically explainable. The most important difference — the IPv6 coverage gap of 6.7 percentage points — is attributable to weighting methodology rather than any fundamental disagreement in the data.

One limitation of this reproduction is the inability to verify the business-category analysis (Table 2 in the paper), as this requires additional organizational classification data not included in the public dataset.

8. Conclusion

This reproduction project successfully verified the three major quantitative claims of the paper ru-RPKI-ready: the Road Left to Full ROA Adoption. The reproduced results closely match the original paper's findings, with all differences explainable by well-understood methodological factors such as prefix-count vs address-space weighting and minor dataset snapshot timing differences.

The paper's core message is confirmed: RPKI adoption has grown significantly but remains incomplete, and a large proportion of the remaining gap consists of Low-Hanging Fruit — prefixes that organizations with existing RPKI infrastructure could protect quickly. The ru-RPKI-ready dataset and platform provide a valuable tool for the networking community to prioritize and accelerate RPKI deployment.

This reproduction demonstrates the value of open datasets in networking research. The availability of the pre-processed Parquet dataset made it possible to reproduce the paper's key findings without needing to rebuild the entire data pipeline from raw BGP dumps. The authors are commended for making their dataset publicly available in a well-structured, queryable format.

9. References

- Deepak Gouda, Romain Fontugne, Cecilia Testart. ru-RPKI-ready: the Road Left to Full ROA Adoption. ACM Internet Measurement Conference (IMC 2025), Madison, WI, USA, October 28-31, 2025. DOI: 10.1145/3730567.3764452
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